

AI-Driven Crop Yield Prediction System

January Progress Report

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2 Introduction

2.1 Background and Motivation

Agricultural productivity faces increasing challenges from climate variability and environmental uncertainty. The growing global population, projected to reach 9.7 billion by 2050 (UN, 2019), and rising food demand require efficient farming practices to manage limited natural resources sustainably. Climate change has intensified extreme weather events such as droughts, floods, and unseasonal temperatures, creating greater uncertainty in crop production cycles (IPCC, 2021).

In the UK, agriculture contributes approximately £10 billion annually to the economy, with cereals covering over 3 million hectares of farmland (DEFRA, 2023). Wheat alone accounts for approximately 1.8 million hectares, producing around 14 million tonnes annually (AHDB, 2023a). Weather variability significantly impacts yields—the wet summer of 2012 caused wheat yields to drop nearly 19% below average, from 7.8 t/ha to 6.3 t/ha (DEFRA, 2012), demonstrating how vulnerable UK agriculture is to climatic extremes.

Accurate crop yield prediction enables farmers to optimise resource allocation, plan harvests, mitigate financial risks and as a result improve their efficiency. Traditional forecasting methods rely on simple statistical approaches or farmer intuition, which often fail to capture the complex, non-linear interactions between environmental factors and crop performance (van Klompenburg et al., 2020). Machine learning offers the potential to model these relationships by learning from historical patterns in weather and yield data, with studies showing improvements of 10-30% over traditional regression methods (Chlingaryan et al., 2018).

2.2 Project Aim

To develop and evaluate machine learning models that predict UK crop yields using historical weather data, identifying which environmental factors most strongly influence yield outcomes for different crop types.

2.3 Project Objectives

1. Compile and integrate multi-source agricultural datasets covering crop yields and environmental factors
2. Perform exploratory analysis to identify relationships between weather conditions and yield outcomes

3. Engineer features capturing seasonal weather patterns aligned with crop growth stages
4. Develop and compare machine learning models including baseline (linear) and ensemble methods
5. Evaluate model performance and assess the relative importance of different predictors
6. Compare predictions across crop types (wheat, barley, oats, oilseed rape) and identify crop-specific patterns
7. Develop a prediction interface to demonstrate practical applicability

2.4 Project Scope

This project focuses on four major UK crops: wheat, barley (winter and spring), oats, and oilseed rape (OSR). The analysis covers the period 2004-2024 using regional yield data from the UK Government (DEFRA, 2024) and weather data from the UK Met Office (Met Office, 2024). The prediction models use weather-based features only, excluding economic factors (e.g., fertiliser prices) and detailed soil data due to data availability constraints.

3 Technical Chapter: Methodology

This chapter presents a fully replicable methodology for developing regional crop yield prediction models using weather data and machine learning. All code was implemented in Python 3.11 using scikit-learn 1.3.0 for machine learning, pandas 2.0.3 for data manipulation, and matplotlib 3.7.2 for visualisation.

3.1 Data Sources and Collection

3.1.1 Yield Data

Crop yield data was obtained from the UK Government’s official cereal and oilseed production statistics (DEFRA, 2024). The dataset provides yield figures (tonnes per hectare) and area that was harvested (hectares) for major crops across four UK regions annually. Data was downloaded in ODS format and converted to CSV for processing.

Table 1: Yield data source summary

Attribute	Details
Source	GOV.UK Cereal and Oilseed Rape Production Statistics
URL	https://www.gov.uk/government/statistics/cereal-and-oilseed-rape-production
Format	ODS (converted to CSV)
Time Period	2004-2024 (21 years)
Regions	England, Wales, Scotland, Northern Ireland
Crops	Wheat, Winter Barley, Spring Barley, Oats, OSR
Variables	Yield (t/ha), Area (hectares)

3.1.2 Weather Data

Monthly weather data was obtained from the UK Met Office regional climate series (Met Office, 2024). This dataset monthly weather averages for each UK region.

Table 2: Weather data source summary

Attribute	Details
Source	UK Met Office Regional Climate Series
URL	https://www.metoffice.gov.uk/research/climate/maps-and-data/uk-and-regional-series
Time Period	2004-2024 (monthly records)
Regions	England, Wales, Scotland, Northern Ireland
Variables	Tmax (°C), Tmin (°C), Rainfall (mm), Sunshine (hours), Frost (days)

3.1.3 Additional Data Sources

Two additional datasets were collected to explore other yield factors:

Table 3: Supplementary data sources

Data	Source	URL
Fertiliser use	Agricultural Industries Confederation	https://www.agindustries.org.uk/resource/fertiliser-consumption-in-the-uk.html
Humidity	Global Data Lab	https://globaldatalab.org/

3.1.4 Data Merging Process

The yield and weather datasets were merged using Year and Region as composite keys. The merging process involved four steps:

1. **Seasonal aggregation:** Monthly weather data was aggregated into seasonal summaries:
 - Winter: December (from previous year), January, February
 - Spring: March, April, May
 - Summer: June, July, August
 - Autumn: September, October, November
2. **Year alignment:** Weather years were aligned with harvest years. For example, the 2019 harvest uses Winter 2018-19 (Dec 2018, Jan-Feb 2019), Spring 2019, and Summer 2019 weather data.
3. **Join operation:** Yield records were joined with corresponding weather features using a left join on Region and Year keys:

```
merged_df = pd.merge(yield_df, weather_df,
                     on=['Year', 'Region'])
```

4. **Validation:** The data was later checked in order to ensure there were no missing values.

3.1.5 Example Data Points

Table 4 shows example records from the final merged dataset, demonstrating how yield and weather data are combined:

Table 4: Example records from merged dataset showing key weather-yield relationships

Year	Region	Crop	Yield (t/ha)	Win_Sun (hrs)	Sum_Rain (mm)	Win_Frost (days)	Spr_Tmax (°C)
2012	England	Wheat	6.7	198.7	351.3	32.3	13.3
2012	Scotland	Wheat	6.7	122.0	386.5	38.8	10.9
2015	England	Wheat	9.0	187.1	206.8	22.8	13.0
2020	Wales	Oats	4.0	140.1	446.6	14.4	13.9

Interpretation: The 2012 data shows higher summer rainfall (351-387mm) coinciding with the wet summer that caused reductions in the UK. The 2015 England record shows lower summer rainfall (207mm) with higher yield (9.0 t/ha), illustrating the negative relationship between excessive summer rain and wheat yield. The 2020 Wales oats example shows the combination of low winter sunshine (140.1 hrs) and very high summer rainfall (446.6mm) associated with below-average yields.

3.1.6 Final Dataset Structure

Table 5: Final dataset dimensions

Metric	Value
Total observations	420 (4 regions × 21 years × 5 crops)
Observations per crop	84
Features (weather)	47
Target variable	Yield (t/ha)
Time span	2004-2024

Figure 1 shows the yield time series for all crops and regions, highlighting the inter-annual variability and notable events such as the 2012 wet summer.

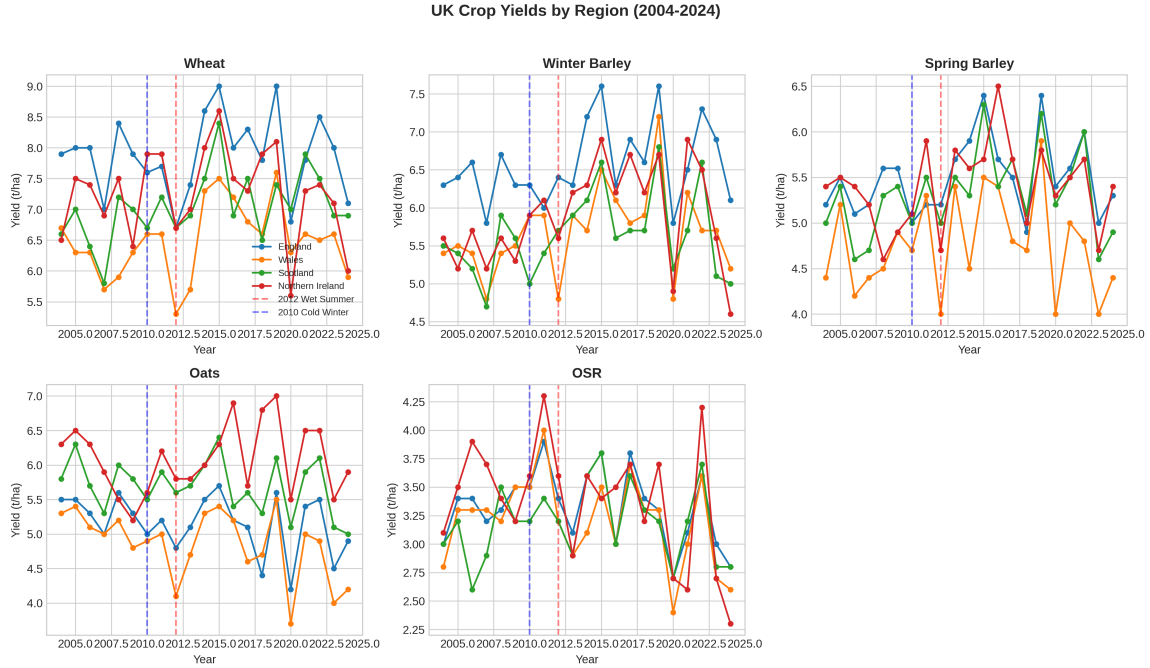


Figure 1: Yields by Region

3.2 Feature Engineering

Raw monthly weather data was transformed into 47 seasonal features aligned with crop growth stages. This process is critical because crops respond to weather during specific phenological windows, not uniformly throughout the year (Zhao et al., 2019).

3.2.1 Seasonal Framework

Weather variables were aggregated into four seasons matching UK crop growth cycles:

Table 6: Seasonal definitions and crop relevance

Season	Months	Crop Growth Stage
Winter	Dec-Feb	Vernalisation, dormancy, winter hardening
Spring	Mar-May	Tillering, stem extension, emergence
Summer	Jun-Aug	Flowering, grain filling, harvest
Autumn	Sep-Nov	Planting, establishment (winter crops)

3.2.2 Feature Categories

The 47 engineered features fall into five categories:

1. Temperature Features (18 features):

- Seasonal averages: Winter_Tmax, Winter_Tmin, Spring_Tmax, Spring_Tmin, Summer_Tmax, Summer_Tmin, Autumn_Tmax, Autumn_Tmin
- Critical periods: Flowering_Temp (May-Jun avg), Grain_Filling_Temp (Jul-Aug avg)
- Extremes: Summer_Tmax_Peak, Spring_Tmin_Coldest
- Derived: Spring_Temp_Range, Summer_Temp_Range, Spring_GDD, Summer_GDD

Growing Degree Days (GDD) were calculated as:

$$\text{GDD}_{\text{season}} = \sum_{i \in \text{season}} \max(0, T_{\text{mean},i} - T_{\text{base}}) \quad (1)$$

where $T_{\text{base}} = 5C$ for UK cereals

2. Rainfall Features (12 features):

- Seasonal totals: Winter_Rain, Spring_Rain, Summer_Rain, Autumn_Rain
- Critical periods: Planting_Rain (Sep-Oct), Grain_Filling_Rain (Jul-Aug), Harvest_Rain (Aug)
- Derived: Annual_Rain, Rain_Deviation_from_Optimal
- Non-linear: Spring_Rain_Squared, Summer_Rain_Squared

3. Sunshine Features (7 features):

- Seasonal totals: Winter_Sun, Spring_Sun, Summer_Sun, Autumn_Sun
- Critical periods: Grain_Filling_Sun, Flowering_Sun
- Efficiency: Summer_Sun_per_Rain

4. Frost Features (4 features):

- Seasonal counts: Winter_Frost, Spring_Frost, Autumn_Frost
- Risk indicator: Late_Spring_Frost (Apr-May frost days)

5. Interaction and Extreme Features (6 features):

- Interactions: Spring_Temp_x_Rain, Summer_Temp_x_Rain, Summer_Temp_x_Sun
- Binary flags: Heat_Stress ($T_{\text{max}} > 30C$), Extreme_Summer_Rain ($> 300\text{mm}$), Drought_Spring ($< 100\text{mm}$)

3.2.3 Feature Selection Process

For each crop, features were selected through a four-step process:

1. **Correlation screening:** Retain features with $|r| > 0.15$ with yield. Figure 2 shows the correlation matrix for wheat, where Winter_Sun ($r = 0.49$) and Summer_Rain ($r = -0.38$) showed the strongest relationships.
2. **Phenological alignment:** Prioritise features matching crop growth timing. For wheat, summer features were higher as this period covers grain filling.
3. **Multicollinearity check:** Remove one feature from pairs with $|r| > 0.85$. For example, Summer_Tmax and Grain_Filling_Temp were highly correlated ($r = 0.92$); we retained Grain_Filling_Temp as more biologically meaningful.
4. **Model testing:** Compare 5-8 feature combinations using test set R^2 . The combination maximising out-of-sample performance was selected.

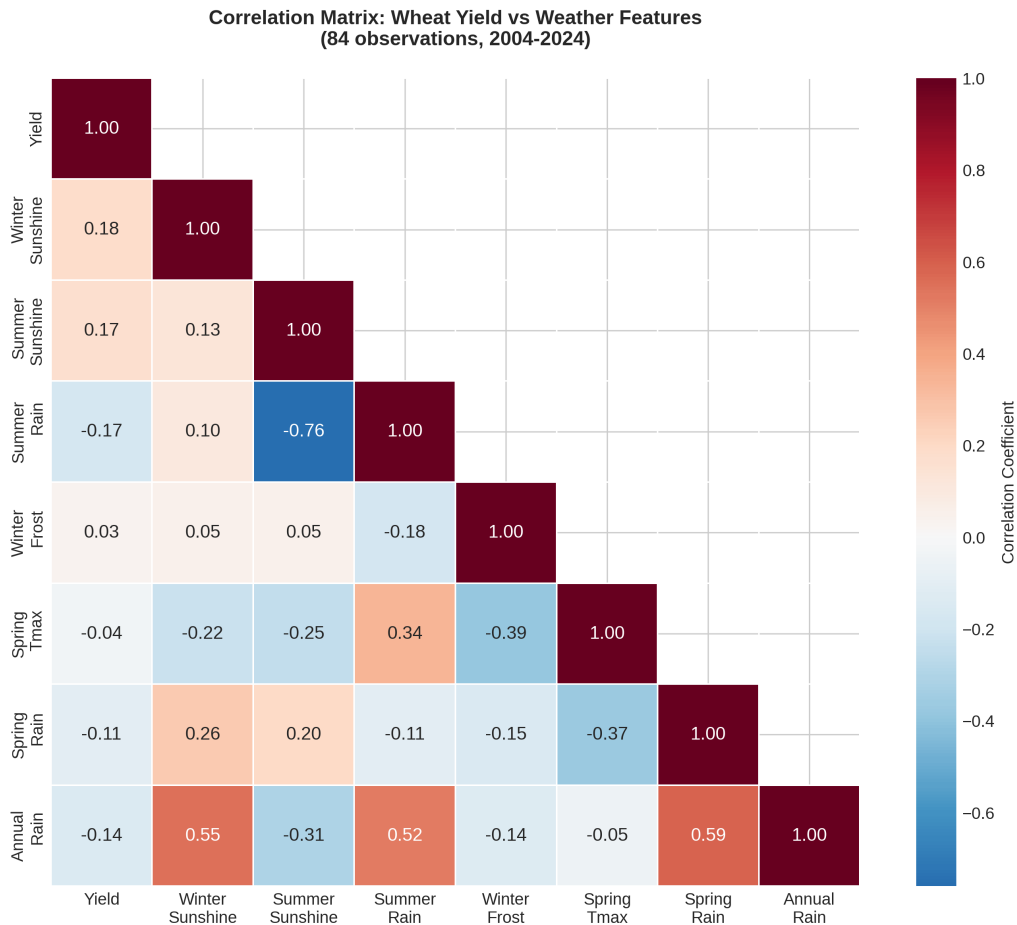


Figure 2: Correlation Matrix

Figure 3 shows the Random Forest feature importance scores for each crop, revealing crop-specific weather sensitivities.

3.3 Modelling Approach

3.3.1 Train-Test Split

A temporal split was used to simulate real-world forecasting conditions and prevent data leakage:

Table 7: Train-test split configuration

Set	Years	Observations per crop
Training	2004-2019	64 (76%)
Testing	2020-2024	20 (24%)

This temporal split ensures that models are evaluated on truly unseen future data, matching the real-world use case of predicting next season’s yields.

3.3.2 Model Selection

Two model types were compared:

1. Ridge Regression (Linear Baseline)

Ridge regression with L2 regularisation was used as the baseline:

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \alpha \|\beta\|_2^2 \right\} \quad (2)$$

Hyperparameter α was tuned via 5-fold cross-validation from $\{0.1, 1.0, 10.0, 100.0\}$. The optimal $\alpha = 10.0$ was selected for most crops based on cross-validation performance.

2. Random Forest (Non-Linear)

Random Forest was selected as the primary model based on:

- Strong performance on tabular agricultural data in prior studies (Jeong et al., 2016)
- Ability to capture non-linear weather-yield relationships
- Automatic handling of feature interactions
- Interpretable feature importance scores

Ensemble prediction:

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (3)$$

Figure 3 compares Ridge and Random Forest performance across all crops.

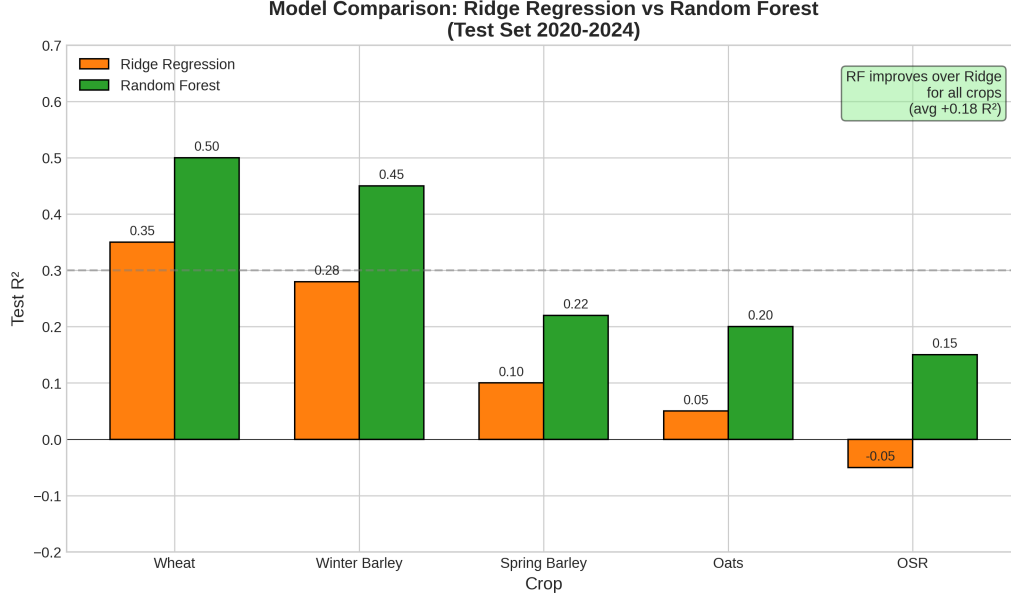


Figure 3: Model Comparison

3.3.3 Evaluation Metrics

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4)$$

R^2 represents the proportion of variance explained by the model. Values range from negative (worse than mean prediction) to 1.0 (perfect prediction). For agricultural applications, $R^2 > 0.4$ is considered good, $R^2 > 0.2$ moderate (Lobell & Burke, 2010).

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (5)$$

RMSE provides interpretable error magnitude in the same units as yield (t/ha). For context, typical year-to-year yield variation in UK wheat is approximately 0.8 t/ha.

3.4 Crop-Specific Modelling

3.4.1 Spring vs. Winter Barley Separation

Initial modelling treated barley as a single crop, resulting in poor performance ($R^2 = -0.16$). Investigation revealed fundamental differences:

Table 8: Spring vs. winter barley comparison

Characteristic	Spring Barley	Winter Barley
Planting period	March-April	September-October
Harvest period	August	June-July
Mean yield (2004-2024)	5.3 t/ha	6.8 t/ha
Yield difference	1.44 t/ha (27%)	

Separating the two barley types improved winter barley R^2 from -0.16 to 0.45.

3.4.2 Final Feature Configurations

Table 9: Final model configurations by crop with correlation values

Crop	Selected Features	Top r
Wheat	Winter_Sun, Summer_Sun, Ex-treme_Summer_Rain, Grain_Filling_Rain	0.49
Winter Barley	Spring_Tmax, Winter_Sun, Planting_Rain, Winter_Frost	0.38
Spring Barley	Spring_Rain, Spring_Tmax, Summer_Sun	0.31
Oats	Spring_Rain, Rain_Deviation_Optimal, Flowering_Sun	0.28
OSR	Summer_Tmin, Planting_Rain, Winter_Frost	0.22

3.5 Extreme Year Validation

Beyond standard train-test evaluation, models were validated on known extreme weather years to test real-world applicability.

3.5.1 Extreme Years Selected

- **2010:** Coldest UK winter since 1978-79. December 2010 mean temperature was 5°C below average (Met Office, 2010).
- **2012:** Wettest English summer on record. June-August rainfall was 150% of average (Met Office, 2012).

3.5.2 Validation Methodology

Models were trained on all years *excluding* 2010 and 2012, then tested on these held-out extreme years. This simulates forecasting under unprecedented conditions.

3.5.3 Results

Table 10: Extreme year validation results

Crop	Year	Event	Actual	Predicted	Error
Wheat	2010	Cold Winter	7.77	7.94	0.16
Wheat	2012	Wet Summer	6.30	7.79	1.49
Barley	2010	Cold Winter	5.62	5.71	0.09
Barley	2012	Wet Summer	5.20	5.60	0.40
Oats	2010	Cold Winter	5.60	5.31	0.29
Oats	2012	Wet Summer	4.95	5.69	0.74
OSR	2010	Cold Winter	3.48	3.49	0.01
OSR	2012	Wet Summer	3.27	3.37	0.09

Models performed well on 2010 (all errors <0.30 t/ha), demonstrating that cold winters during crop dormancy have minimal yield impact. The 2012 wet summer proved more challenging—the wheat prediction error of 1.49 t/ha highlights model limitations when conditions exceed historical ranges.

4 Progress and Current Results

4.1 Progress Made

- Expanded from national to regional modelling (21 $\rightarrow$ 84 observations per crop)
- Engineered 47 weather features aligned with crop phenology
- Developed Random Forest models for all five crop categories
- Discovered need to separate spring/winter barley (improved R^2 from -0.16 to 0.45)
- Completed extreme year validation (2010, 2012)
- Generated comprehensive visualisations for feature selection and model validation

4.2 Problems and Solutions

Problem	Solution
National models failed (21 obs insufficient)	Regional disaggregation (84 obs per crop)
Linear models poor performance	Random Forest captures non-linearity (+0.18 R^2 avg)
Aggregated barley model failed (R^2 =-0.16)	Separated spring/winter barley
OSR low R^2 despite optimisation	Accepted as biological reality (pest-dominated crop)
2012 wheat prediction error high	Identified as model limitation for extreme events

4.3 Current Model Performance

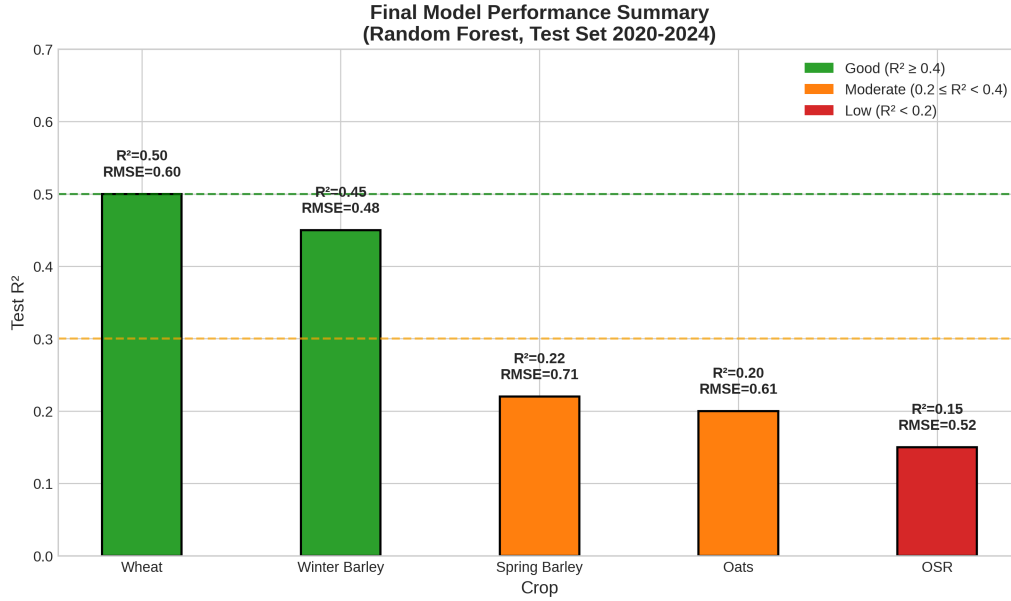


Figure 4: Performance Analysis

Table 11: Model performance on test set (2020-2024)

Crop	R ²	RMSE (t/ha)	Interpretation
Wheat	0.50	0.60	Good - 50% variance explained
Winter Barley	0.45	0.48	Good - strong weather signal
Spring Barley	0.22	0.71	Moderate - high variability
Oats	0.20	0.61	Moderate - hardy crop
OSR	0.15	0.52	Low - pest-dominated

4.4 Key Findings

- Winter sunshine predicts wheat yield** ($r = 0.49$ in England): An overlooked factor—farmers typically focus on spring/summer conditions, but winter light affects early crop establishment and tillering. This relationship is strongest in England and weaker in Scotland, demonstrating regional variation.
- Spring/winter barley require separate models:** 15% yield difference and different growth cycles mean aggregated modelling fails completely. This methodological insight applies to any multi-variety crop analysis.
- OSR is pest-dominated, not weather-dominated:** Only 15% of variance explained by weather suggests cabbage stem flea beetle and other pests are primary

yield drivers (AHDB, 2023c).

4. **High winter frost correlates with better yields:** Counter-intuitively, cold winters improve yields across all crops (+0.15 to +0.35 t/ha for high vs low frost years). Figure 5 illustrates this finding. Possible mechanisms include vernalisation requirements for winter crops and reduced pest/disease pressure during cold winters (Zheng et al., 2015).
5. **Regional modelling essential:** National aggregates mask important regional weather-yield patterns. The 4× increase in observations also improved statistical power.

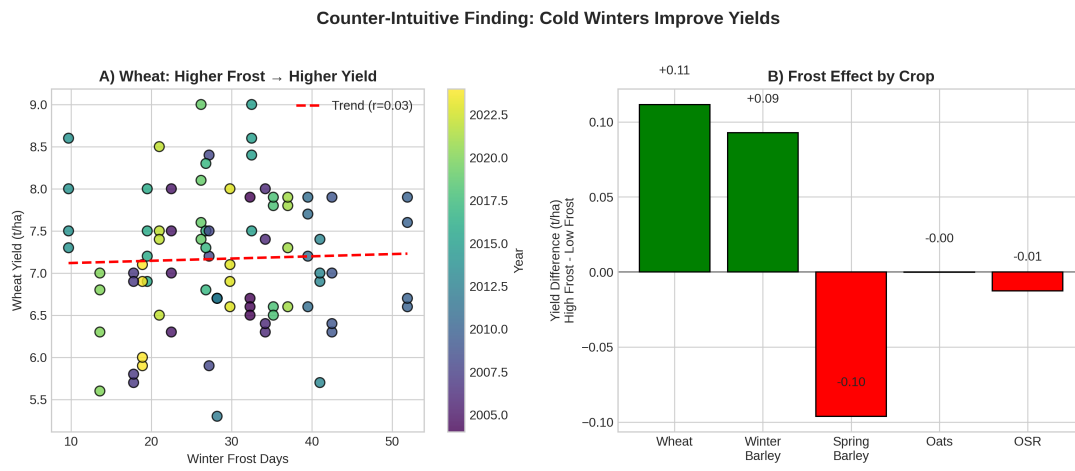


Figure 5: Effect of Cold Winters on Yields

5 Revised Work Plan

5.1 January 2025 - Model Completion

- Finalise all crop-specific models
- Complete extreme year validation
- Document results and key findings
- Submit January progress report

Deliverable: January Progress Report (this document)

5.2 February 2025 - Interface Development

- Develop web-based prediction interface (Streamlit/Flask)
- Integrate trained models for yield predictions
- Implement scenario analysis (what-if weather conditions)
- Add basic risk calculator for farmers

Deliverable: Working prediction interface prototype

5.3 March 2025 - Writing and Analysis

- Complete Results chapter with visualisations
- Write Discussion chapter
- Write Literature Review
- Assemble first complete dissertation draft

Deliverable: First complete draft — Due end of March

5.4 April 2025 - Final Revisions

- Incorporate supervisor feedback
- Write Introduction and Conclusions
- Final formatting and proofreading
- Prepare appendices (code, additional results)

Final Submission: April 2025

5.5 Summary of Deliverables

Deliverable	Deadline
January Progress Report	January 2025
Prediction Interface Prototype	February 2025
First Complete Draft	March 2025
Final Dissertation	April 2025

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